

APRS COMMAND

User Manual

Complete Guide to Installation, Setup, and Operation

A free, open-source APRS client for amateur radio operators.

macOS · Windows 11 · Linux · Raspberry Pi

Developer

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Version

1.1 · Updated July 27, 2026 · Applies to v0.5.0-alpha

License

GNU General Public License v3

Repository

github.com/KE4CON/APRS-Command

Supported Platforms

- macOS (Apple Silicon & Intel)
- Windows 11 (64-bit)
- Linux (Ubuntu, Mint, Debian, and others)
- Raspberry Pi 4 / 5 (64-bit OS)

Status

Alpha — functional for daily use, active development

Built on the vision of Bob Bruninga WB4APR · In memory of Roger Barker G4IDE

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Chapter 1

Introduction

1.1 The Vision Behind APRS Command

APRS Command is built on the original vision of Bob Bruninga WB4APR, the father of APRS. Bob conceived APRS in the early 1990s not as a tracking system, but as a situational awareness tool — a way for amateur radio operators to share real-time tactical information about what is happening in an area. Weather, resources, messages, objects, coverage — a common operating picture for emergency communications and public service events.

That vision guides every design decision in this program.

APRS Command is also inspired by Roger Barker G4IDE, author of UI-View32, a widely used APRS client that many operators came to rely on heavily. When Roger passed in 2004, the source code was destroyed per his wishes, and UI-View32 could never be updated, fixed, or ported to new platforms. A program that many operators depended on was frozen in time and slowly left behind by the operating systems it ran on.

APRS Command is released under GPL v3 so that it can live on and be improved by the community indefinitely. No one can ever take it closed source. If the original author is gone tomorrow, any amateur radio operator in the world can pick it up and carry it forward.

1.2 What Is APRS?

APRS stands for Automatic Packet Reporting System. It is a real-time digital communication system used by amateur radio operators worldwide. Unlike a simple radio conversation, APRS lets stations broadcast information — position, weather, messages, and status — to anyone listening in the area, and to the entire world through a network of internet-connected receivers.

The APRS network operates on 144.390 MHz in North America. Every transmission is a brief digital packet containing structured data — a position, a weather reading, a message, an object. Together these packets paint a live picture of what is happening in an area.

**Tip**

APRS requires a valid amateur radio license to transmit. You do not need a license to receive and view APRS data. Contact your national amateur radio organization for licensing information.

1.3 What Is APRS Command?

APRS Command is a free, open-source APRS client for macOS, Windows 11, Linux, and Raspberry Pi — all from the same codebase. It displays APRS stations on a live interactive map, lets you send and receive messages, manage nets, predict radio coverage, track weather data, and export standard ICS emergency management documentation.

Written in C# and .NET 10 using the Avalonia UI framework. Licensed under GPL v3. Source code: github.com/KE4CON/APRS-Command.

1.4 How APRS Command Connects to the World

APRS Command connects to the APRS network in two ways:

- APRS-IS (Internet Service) — connects through your internet connection to a global network of servers. Receive packets from your area and, with a valid passcode, transmit your position and messages. Easiest starting point for most operators.
- RF (Radio Frequency) — with a radio and TNC, send and receive packets directly over the air without an internet connection. Ideal for field operations, emergency communications, and iGate operation.



Tip

For most beginners, APRS-IS is the easiest starting point. You can get on the air and see stations on the map within minutes using only your internet connection.

1.5 APRS-IS Passcodes

To transmit on APRS-IS, you need a passcode — a number calculated from your callsign. It never changes. Find yours at apps.magicbug.co.uk/passcode. Keep it private.

Without a passcode (using -1), you can still receive all APRS data and see stations on the map — you simply cannot transmit.



Important

If you do not yet have a passcode, enter -1 in the Passcode field. The app will connect and receive data normally — you simply will not be able to transmit until you enter your real passcode.

1.6 Who This Manual Is For

This manual is written for everyone — whether you have used APRS for decades or this is your first encounter. No prior APRS knowledge is assumed. Experienced operators may skim the conceptual sections and jump to the features they need.

Chapter 2

Installation

2.1 What You Are Installing

APRS Command ships two ways on every platform: a native installer (recommended for most people) and a portable archive (zip or tar.gz) that runs from any folder with no installation step at all. The installer places APRS Command in the normal application location for your platform and adds Start Menu / Desktop / Applications entries; the portable archive is entirely self-contained and leaves nothing outside the folder you extract it to. Either way, your settings are stored separately in your user profile and are unaffected by reinstalling, upgrading, or moving the app folder. No administrator account or special permissions are required to run APRS Command — only the Windows installer itself needs administrator rights to write to Program Files.

The application executable inside the zip is named Aprs.Desktop (macOS and Linux) or Aprs.Desktop.exe (Windows). The folder also contains supporting libraries required to run the application — do not delete individual files from the folder.

Important

APRS Command is not code-signed. Every platform will show a security warning on first launch. This is expected. The steps below explain exactly how to proceed on each platform.

2.2 System Requirements

Platform	Minimum Requirement
macOS	macOS 12 Monterey or later. Apple Silicon (M1/M2/M3/M4) or Intel. 4 GB RAM.
Windows 11	Windows 10 or 11, 64-bit. 4 GB RAM.
Linux (64-bit)	Ubuntu 20.04+, Linux Mint 21+, Debian 11+, Fedora 36+, or any modern 64-bit distro. 4 GB RAM.
Raspberry Pi	Raspberry Pi 4 or 5. 4 GB RAM recommended. Raspberry Pi OS 64-bit (not 32-bit).

2.3 Downloading APRS Command

Visit github.com/KE4CON/APRS-Command/releases and download the file matching your platform. Each release lists both an installer (.dmg, -Setup.exe, .deb, or .rpm) and a portable archive (.zip or .tar.gz) for every platform — the table below shows both. The filename includes the version number and platform name.

[Screenshot]

GitHub releases page showing all download options with filenames

Platform	File to Download
macOS — Apple Silicon (M1/M2/M3/M4)	Installer: APRSCommand-vX.X.X-macos-arm64.dmg · Portable: APRSCommand-vX.X.X-macos-arm64.zip
macOS — Intel	Installer: APRSCommand-vX.X.X-macos-x64.dmg · Portable: APRSCommand-vX.X.X-macos-x64.zip
Windows 10 / 11 (64-bit)	Installer: APRSCommand-vX.X.X-windows-x64-Setup.exe · Portable: APRSCommand-vX.X.X-windows-x64.zip
Linux (64-bit desktop/laptop/server)	Installer: aprs-command_X.Y.Z_amd64.deb (or .rpm) · Portable: APRSCommand-vX.X.X-linux-x64.tar.gz
Raspberry Pi (ARM64)	Installer: aprs-command_X.Y.Z_arm64.deb (or .rpm) · Portable: APRSCommand-vX.X.X-linux-arm64.tar.gz

● Important

Raspberry Pi: download the ARM64 build (filenames containing -arm64, e.g. aprs-command_X.Y.Z_arm64.deb or APRSCommand-vX.Y.Z-linux-arm64.tar.gz) — not the Linux x64 version. The x64 version targets standard PC processors and will not run on Pi hardware.

If you are unsure which Mac you have: Apple menu > About This Mac. "Apple M1/M2/M3/M4" = Apple Silicon (arm64). "Intel Core" = Intel (x64).

2.4 Installing on macOS

APRS Command offers a standard .dmg installer for macOS, plus a portable zip for anyone who prefers to run the app directly from a folder with no installation step. Neither path is code-signed with an Apple developer certificate, so macOS Gatekeeper will block first launch either way — the steps below clear that block for whichever method you choose.

Installer Method (Recommended)

1. Download APRSCommand-vX.Y.Z-macos-arm64.dmg (Apple Silicon M1/M2/M3/M4) or APRSCommand-vX.Y.Z-macos-x64.dmg (Intel Mac).
2. Double-click the .dmg file to mount it.
3. Drag APRS Command.app into your Applications folder.
4. First launch only: in Applications, right-click (or Control-click) APRS Command, choose Open, then click Open again in the confirmation dialog. macOS remembers this exception — every launch after the first works normally by double-clicking.

Portable Archive Method (Alternative)

Step 1 — Unzip

1. Double-click the downloaded zip file. macOS unzips it and creates a folder named after the zip file in the same location.
2. Move the unzipped folder to wherever you want to keep it — your home folder, Applications, or a dedicated tools folder. The app runs from wherever you put it.

Step 2 — Clear the Gatekeeper quarantine

Because the app is unsigned, macOS quarantines every file inside the folder. The most reliable way to clear this is with a single Terminal command:

```
xattr -cr /path/to/APRSCommand-vX.X.X-macos-arm64/
```

Replace the path with the actual path to your unzipped folder. The easiest way: type `xattr -cr` followed by a space, then drag the folder from Finder into the Terminal window — macOS fills in the path automatically. Press Enter.

3. Open the folder in Finder and double-click `Aprs.Desktop` to launch.



Tip

Alternatively, right-click `Aprs.Desktop` and choose Open. macOS will ask you to confirm. Click Open. This works for the first launch but only clears the quarantine on that one file — `xattr -cr` on the folder is cleaner because it clears all files at once.

[Screenshot]

Terminal showing xattr -cr command and macOS Finder with Aprs.Desktop highlighted



Note

You only need to do this once. After the quarantine is cleared, `Aprs.Desktop` opens normally by double-clicking every time after.

2.5 Installing on Windows 11

APRS Command offers a standard `Setup.exe` installer for Windows, plus a portable zip you can run directly with no installation step. Neither path is signed with a code-signing certificate, so Windows SmartScreen will warn on first launch either way — the steps below show what to expect and how to proceed for whichever method you choose.

Installer Method (Recommended)

1. Download `APRSCommand-vX.Y.Z-windows-x64-Setup.exe`.
2. Double-click to run it.
3. First launch only: if SmartScreen shows "Windows protected your PC," click More info, then Run anyway.
4. APRS Command installs to `C:\Program Files\APRS Command` with Start Menu and Desktop shortcuts.
5. To uninstall later: Settings → Apps → APRS Command.

Portable Archive Method (Alternative)

1. Right-click the downloaded zip file and choose Extract All. Choose a destination folder — your Documents folder or a dedicated tools folder works well.
2. Open the extracted folder. You will see Aprs.Desktop.exe along with a number of .dll files. Do not delete any of these files.
3. Double-click Aprs.Desktop.exe. Windows SmartScreen shows a blue banner: "Windows protected your PC."
4. Click More info. A "Run anyway" button appears at the bottom of the dialog.
5. Click Run anyway. The app launches.

[Screenshot]

Windows SmartScreen dialog — "More info" and "Run anyway" buttons highlighted



Note

You only need to click through SmartScreen once per executable. Windows remembers your approval. After the first launch, double-clicking Aprs.Desktop.exe opens the app immediately.



Tip

To make APRS Command easy to find: right-click Aprs.Desktop.exe and choose Send to > Desktop (create shortcut), or Pin to Start. The shortcut can be renamed to "APRS Command" without affecting the application.



Important

If Windows Firewall asks whether to allow network access, click Allow access. APRS Command requires network access to connect to APRS-IS. If you accidentally click Block, go to Windows Defender Firewall > Allow an app through the firewall and add Aprs.Desktop.exe manually.

2.6 Installing on Linux

APRS Command provides native .deb packages (Debian, Ubuntu, Linux Mint, Raspberry Pi OS, and derivatives) and .rpm packages (Fedora, RHEL, CentOS, openSUSE, and derivatives), plus a portable tar.gz archive that runs on any 64-bit Linux distribution with no package manager interaction or sudo access required at all.

Installer Method (Recommended)

1. Debian, Ubuntu, Linux Mint, Raspberry Pi OS, and derivatives: `sudo dpkg -i aprs-command_X.Y.Z_amd64.deb`
2. Fedora, RHEL, CentOS, openSUSE, and derivatives: `sudo rpm -i aprs-command-X.Y.Z-1.x86_64.rpm`
3. APRS Command installs to /opt/aprs-command/ with a launcher at /usr/local/bin/aprs-command and a .desktop entry added to your application menu automatically.
4. Launch from a terminal (aprs-command) or from your application menu.

Portable Archive Method (Alternative)

1. Extract the download: `tar -xzf APRSCommand-vX.X.X-linux-x64.tar.gz`

2. Enter the extracted folder: `cd APRS-Command-linux-x64`
3. Make the executable runnable: `chmod +x Aprs.Desktop`
4. Launch: `./Aprs.Desktop`

 Tip

To add APRS Command to your application menu, create a .desktop launcher file in `~/.local/share/applications/`. See the project wiki on GitHub for the exact file contents including the correct Exec path and icon.

2.7 Installing on Raspberry Pi

Installation follows the same installer-or-portable choice as Linux (Section 2.6), using the ARM64 build. The .deb installer is recommended — it registers APRS Command in your application menu automatically, which is convenient if you're running the Pi with a display attached. Run these commands in a terminal on the Pi:

Installer Method (Recommended)

1. Download `aprs-command_X.Y.Z_arm64.deb` from the releases page.
2. Install it: `sudo dpkg -i aprs-command_X.Y.Z_arm64.deb`
3. Launch from a terminal (`aprs-command`) or from the application menu if running with a display attached.

Portable Archive Method (Alternative)

1. Download the `-linux-arm64.tar.gz` file from the releases page.
2. Extract: `tar -xzf APRSCommand-vX.X.X-linux-arm64.tar.gz`
3. Enter the folder: `cd APRS-Command-linux-arm64`
4. Make executable: `chmod +x Aprs.Desktop`
5. Launch: `./Aprs.Desktop`

 Note

APRS Command may feel slightly slower on Raspberry Pi 4 than on a desktop when the map contains many stations. This is normal. Pi 5 is noticeably faster.

Chapter 3

First Launch and Setup Wizard

3.1 What Happens on First Launch

The very first time APRS Command opens, a setup wizard appears to collect the minimum information needed: your callsign and your location. Everything else is configured later through Settings. The app does not transmit anything until you explicitly enable it — receive-only mode is the safe default.

[Screenshot]

Setup wizard — welcome screen

3.2 Step 1 — Your Callsign

Enter your amateur radio callsign exactly as issued by your licensing authority. APRS Command automatically converts it to uppercase.

Field	What It Means	Example
Callsign	Your amateur radio callsign. Letters and numbers only, no spaces.	KE4CON
SSID	A number 0–15 letting one callsign represent multiple stations. 0 = home station. 1 = mobile or secondary. Leave at 0 if unsure.	1

 Tip

SSID stands for Secondary Station Identifier. KE4CON with SSID 0 is your primary station. KE4CON-1 through KE4CON-15 can represent a vehicle, portable unit, or weather station.

 Important

Your callsign must match your amateur radio license exactly. Transmitting under another person's callsign is illegal (FCC Part 97).

[Screenshot]

Setup wizard — callsign entry screen

3.3 Step 2 — Your Location

APRS Command needs to know your station location to center the map, calculate distances, and position your beacon.

Field	What It Means	Example
Latitude	Decimal degrees. Positive = North. Negative = South.	41.9756
Longitude	Decimal degrees. Negative = West (all of North America). Positive = East.	-88.4553

Tip

To find your coordinates: open Google Maps or Apple Maps, search for your address, and right-click your location. Decimal-degree coordinates appear at the top of the context menu.

Note

Your location is not transmitted unless you explicitly send a beacon. Setting it here only configures the app locally.

[Screenshot]

Setup wizard — location entry with latitude and longitude fields

3.4 Completing the Wizard

Click Finish. APRS Command opens the main window and begins connecting to APRS-IS. Stations will begin appearing on the map within a few seconds.

[Screenshot]

Main window after first setup — map with local stations visible

Chapter 4

Settings

The Settings window contains all configuration options. Open it from Settings in the menu bar. Work through the tabs in order the first time you configure the app.

[Screenshot]
Settings window — all tabs visible

4.1 Station Setup

The most important settings page. Defines who you are on the APRS network and how your station behaves.

[Screenshot]
Settings — Station Setup tab

4.1.1 Callsign and SSID

Field	What It Means	Example
Callsign	Your full amateur radio callsign. Automatically uppercased on save.	KE4CON
SSID	0–15. Home station = 0. Mobile/secondary = 1. Leave at 0 if unsure.	1
Station Comment	Short text appearing alongside your callsign on other operators' maps.	John / 147.195 / Home

4.1.2 Location

Field	What It Means	Example
Latitude	Decimal degrees. North is positive.	41.9756
Longitude	Decimal degrees. West (North America) is negative.	-88.4553
Fixed Station Mode	Check if your station does not move. Uses entered coordinates instead of GPS.	Checked for home stations

4.1.3 Distance Unit Preference

Choose whether distances display in miles or kilometers throughout the entire application. The app converts to kilometers internally wherever APRS-IS requires it — you never need to do the conversion manually.

Setting	When to Choose It
Miles (US / UK)	Based in the United States or United Kingdom. All distances show in miles. This is the default.
Kilometers (metric)	Outside the US/UK and use the metric system. All distances display in kilometers.

4.1.4 Symbol

Your symbol is the icon representing your station on other operators' maps.

Field	What It Means	Example
Symbol Table	/ = primary table (most common symbols). \ = alternate table. Leave as / unless you have a specific reason.	/ (primary)
Symbol Code	Single character selecting your icon. Common: - (house), > (car), * (weather), _ (WX format), r (antenna).	- (house)

Tip

For a home station: Table = / and Code = - displays as a house icon on all APRS maps. A full symbol reference is in Appendix A.

4.1.5 Beacon Settings

A beacon is a periodic transmission announcing your position to the network.

Field	What It Means	Example
Enable Transmit	Master transmit switch. Must be checked for any transmission — beacons or messages.	Checked to enable
Enable APRS-IS Transmit	Controls whether beacons go to APRS-IS. Both this AND Enable Transmit must be checked.	Checked for internet beaconing
APRS-IS Beacon Interval	How often your position is automatically sent, in minutes. Minimum is 5 minutes.	11
Beacon Path	Controls digipeater relaying. Standard: WIDE1-1,WIDE2-1 — no spaces.	WIDE1-1,WIDE2-1

Important

The Beacon Path must never contain spaces. WIDE1-1,WIDE2-1 is correct. Spaces cause APRS-IS to silently

reject your packet — the app reports Transmitted but nothing appears on aprs.fi. APRS Command automatically corrects spaces on save.

4.2 Connections

Configure how APRS Command connects to the APRS network.

[Screenshot]
Settings — Connections tab

4.2.1 APRS-IS Connection

Field	What It Means	Example
Server Host	rotate.aprs2.net auto-selects the best available server. Only change if you have a specific reason.	rotate.aprs2.net
Server Port	Port for APRS-IS. 14580 is standard.	14580
Passcode	Your personal APRS-IS passcode. Required to transmit. Receive-only: use -1. Find yours at apps.magicbug.co.uk/passcode .	-1 or your number
Filter	Geographic radius filter. Built automatically from your Station Setup filter radius. Advanced users can override with a custom filter string.	r/41.97/-88.45/200

Important

Keep your passcode private. Anyone who has it can transmit using your callsign on APRS-IS.

4.3 Audio and Voice Readout

Configure sound alerts for incoming events and voice readout using your operating system's built-in text-to-speech engine. Full Voice Readout documentation is in Section 7.10.

Field	What It Means
Incoming Message Alert	Sound played when a direct APRS message arrives.
Station Alert	Sound played when a tracked station appears or moves.
Geofence Alert	Sound played when a station enters or exits a geofence.
Enable Voice Readout	Master switch for spoken announcements using OS text-to-speech.

Chapter 5

The Main Window

After setup completes, the main window appears. It remembers its size and position between sessions.

[Screenshot]

Main window — fully annotated showing all areas

5.1 Layout Overview

Area	Location	Purpose
Title bar	Top	Program name, callsign, and live connection status indicators.
Menu bar	Below title bar	All features organized into menus.
Left sidebar	Left edge	Icon buttons for the most frequently used tools.
Map area	Center / right	Live interactive map showing all received APRS stations and overlays.

5.2 Title Bar and Status

- Green "APRS-IS Connected" — connected and receiving data.
- Yellow / orange — connecting or reconnecting.
- Red or absent — not connected. Check your internet connection and Connections settings.

[Screenshot]

Title bar close-up — connection status indicators

5.3 Menu Bar Overview

Menu	Contents
Settings	Station setup, connections, audio, and all configuration.
Messages	Compose, send, and read messages. Broadcast, scheduled messages, receipts dashboard.
Net Control	Net roster management, check-in tracking, and function-key shortcuts.
Objects	Create and manage APRS objects — markers for locations and resources.

Events	ICS form export and After-Action Report generation.
Draw	Draw lines, polygons, and circles on the map for planning or briefing.
Weather	Weather station data, NWS alerts, and radar overlay.
Tools	Offline maps, frequency reference, elevation profile, coverage prediction, shadow beacon, net scripts, Winlink gateways, packet statistics, session templates, and more.

5.4 The Sidebar

The left sidebar provides one-click access to frequently used tools. Hover over any icon to see its tooltip. Divided into two groups: map tools at top, operator tools below.

[Screenshot]
Left sidebar — all icons with tooltips visible

5.5 The Map

The map is the heart of APRS Command. All received APRS stations appear as icons on a live interactive map that updates automatically as packets arrive.

- Pan: click and drag with the left mouse button.
- Zoom: scroll wheel up (in) or down (out).
- Select a station: click its icon to see details in the side panel.
- Context menu: right-click anywhere for location-specific options.

[Screenshot]
Map with multiple stations — one selected with details panel open

Chapter 6

Sidebar Tools

Every sidebar icon is documented here in order from top to bottom. Hover over any icon to see its tooltip.

6.1 Home / Reset View

Resets the map view to your station's area. Use this whenever you have panned or zoomed far away and need to get back quickly.

6.2 Center on My Station

Centers the map specifically on your own station's last known position. Useful when tracking your own location with GPS and the map has scrolled away.

6.3 Find Station

Opens a search dialog to locate a specific callsign and center the map on it.

1. Click the search icon.
2. Type the callsign (e.g. W9ABC or W9ABC-1).
3. Click Search or press Enter.
4. If found, the map centers on that station and its details appear in the side panel.

 Tip

Search is case-insensitive. Callsigns must have been heard within your current APRS-IS filter area recently.

6.4 Measure Distance

Click two points on the map to see the straight-line distance and bearing between them.

1. Click the measure icon.
2. Click your starting point on the map or on a station icon.
3. Click your ending point.
4. The result shows distance in your preferred unit and bearing in degrees.

 Note

Distance is a straight-line great-circle distance, not a road distance.

6.5 Map Layer Toggle

Cycles through available base map styles: OpenStreetMap road map, satellite imagery, topographic maps. The current style changes immediately. A tooltip shows the current layer name.

6.6 Beacon Now

Sends your position beacon immediately to APRS-IS, regardless of the next scheduled interval. For a beacon to succeed, all of the following must be true:

- Settings > Station Setup > Enable Transmit: checked
- Settings > Station Setup > Enable APRS-IS Transmit: checked
- Beacon interval: 5 minutes or more
- Valid callsign entered in Station Setup
- Valid passcode entered in Connections (not -1)
- APRS-IS status: green in the title bar

A popup confirms success or explains exactly why the beacon was not sent.

Important

If the popup says Transmitted but nothing appears on aprs.fi within two minutes, check Settings > Station Setup > Beacon Path. The path must have no spaces. WIDE1-1,WIDE2-1 is correct. APRS Command corrects spaces automatically on save — click Save in Station Setup once to trigger the correction.

[Screenshot]

Beacon Now — success confirmation popup

6.7 Alert Status

Opens the alert panel showing active NWS weather alerts, geofence alerts, and system notifications. A badge on the bell icon shows the count of active alerts.

6.8 Range Rings

Draws three concentric range rings centered on your station. The rings are fully transparent so the map remains visible inside them. Click once to show, click again to hide.

Tip

Range rings are useful during net operations to quickly identify which stations fall within a given distance of your location. Right-click the map and choose Set Ring Center to move the rings to any point.

6.9 Station Trails

Toggles movement trails for mobile stations. When enabled, a colored line traces the recent path of any station seen in multiple positions during the current session. Trails are kept for up to 2 hours and up to 30 points per station.

Tip

Trails are most useful for vehicles, aircraft, and other moving stations. Fixed home stations typically have no trail.

6.10 Radar Overlay

Toggles the NEXRAD weather radar overlay on the map. Data is from NOAA and is typically a few minutes behind real time.

1. Click the radar icon (or Weather > Radar Overlay — both do the same thing).
2. After a few seconds, radar imagery appears over the map.
3. Once frames are loaded, an animation scrubber bar appears at the bottom center of the map.
4. Click the icon again to turn radar off.

Tip

The animation scrubber appears a few seconds after turning radar on — it takes time to fetch the frame list from NOAA. This delay is normal.

Important

Radar requires an active internet connection. It is not available for offline or pure RF operation.

[Screenshot]

Map with radar overlay active and animation scrubber at bottom

Chapter 7

Menu Bar Features

This chapter covers every menu item in detail. Features accessible from both the sidebar and the menu bar (such as Radar Overlay) are covered in Chapter 6 and referenced here.

7.1 Settings Menu

Contains all configuration options covered in Chapter 4.

7.2 Messages

7.2.1 Compose Message

Sends a direct APRS message to any station.

1. Open Messages > Compose Message.
2. In the To field, enter the recipient's callsign (e.g. W9ABC-1).
3. Type your message. APRS messages have a 67-character limit.
4. Click Send.

**Tip**

You can also compose a message by clicking a station on the map and choosing Message in its details panel.

**Important**

APRS messages are public on APRS-IS — anyone monitoring can read them. Never send sensitive personal information via APRS messaging.

7.2.2 Message Broadcast

Sends the same message to multiple stations simultaneously.

1. Open Messages > Message Broadcast.
2. Enter callsigns in the recipient list, one per line, or click Import from Roster.
3. Type your message.
4. Click Send to All.

7.2.3 Scheduled Messages

Schedule a message to be sent automatically at a specified future time.

1. Open Messages > Scheduled Messages.
2. Enter the recipient callsign and message text.
3. Set the date and time.
4. Click Schedule.

 Note

Scheduled messages only send if APRS Command is running at the scheduled time. The app checks every 30 seconds.

7.2.4 Message Receipts Dashboard

Shows a summary of all messages sent and received during the current session, including acknowledgment status.

7.3 Net Control

Designed for operators running organized nets — APRS Command helps manage the roster, track station status, and send formatted messages.

[Screenshot]

Net Control window — roster, status buttons, and keyboard shortcuts

7.3.1 The Roster

Lists all stations that have checked in. Each entry shows callsign, check-in status, resource status, resource type, and category.

7.3.2 Status Buttons

- Check In — marks a station as active in the net.
- Standby — marks a station as standing by.
- Departed — marks a station as having left the net.

7.3.3 Keyboard Shortcuts F1-F8

With Net Control open and a station selected, function keys send standard net messages instantly:

Key	Action
F1	Check In selected station
F2	Mark as Standby
F3	Mark as Departed
F4	Send standby acknowledgment
F5	Send acknowledgment message
F6	Mark as Available
F7	Mark as Assigned
F8	Mark as On Scene

Delete	Remove selected station from roster
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7.4 Objects and Area Objects

7.4.1 Creating an Object

APRS objects are markers placed on the map visible to other stations. They represent any location — an incident site, staging area, repeater, or point of interest.

1. Open Objects > New Object, or right-click the map and choose Create Object Here.
2. Enter a name (up to 9 characters — the APRS specification limit).
3. Set the position, or click the map to place it.
4. Choose a symbol.
5. Add a comment if desired.
6. Click Transmit to broadcast to APRS-IS, or Save to keep it locally.

Important

Object names are limited to 9 characters by the APRS specification. Longer names are automatically truncated.

7.4.2 Area Objects

APRS Command can encode a standard APRS area object — a shape (circle, rectangle, triangle, or line) that renders as a drawn region on other APRS clients. Area objects mark zones, coverage areas, incident perimeters, and geographic boundaries.

To create an area object: open the Object Manager, click New Object, and check Area Object. The symbol is automatically set to the APRS area object symbol (\I). Configure:

- Shape — Open/Filled Circle, Rectangle, Triangle, or Line.
- Color — Black, Blue, Green, Cyan, Red, Violet, Amber, or White.
- Width (km) — width or radius in kilometres (0.1–99).
- Height (km) — height in kilometres. Use same as Width for circles.
- Comment — optional description appended after the area encoding.

The encoding is written to the packet comment field per APRS Protocol Reference §11. Other APRS clients that support area objects will render the shape on their map.

Note

Area object visual rendering depends on the receiving client. APRS Command encodes the shape correctly per the APRS specification.

7.5 Draw Tools and Map Import/Export

Draw shapes on the map for planning, briefing, or annotation. Drawings are visible only locally — never transmitted over APRS.

- Draw Line — click two or more points to draw a path. Double-click to finish.
- Draw Polygon — click three or more points to draw a filled area. Double-click to close.

- Draw Circle — click a center point, then drag to set the radius.
- Erase Shape — click a shape to remove it.
- Clear All Drawings — removes every drawn shape at once.

 Tip

Draw tools are excellent for marking a search area, event perimeter, or planned route during pre-activation planning.

7.5.1 Importing GPX and KML Files

APRS Command can import route files, track logs, and waypoint collections from GPS devices, mapping software, and field planning tools. Use File > Map > Import GPX/KML to open a file.

Supported formats:

- GPX (GPS Exchange Format) — the standard format for Garmin devices, Strava, AllTrails, and most GPS units. Tracks, routes, and waypoints are all imported.
- KML (Keyhole Markup Language) — the format used by Google Earth, CalTopo, and many GIS tools. Points, lines, and polygons are imported.

Imported geometry appears in the drawing layer on the map immediately. Tracks appear in APRS blue, routes in gold, and waypoints as labeled dot markers. All imported shapes can be erased using the Erase tool.

 Note

KMZ files (zipped KML) are not directly supported. To import a KMZ file, rename it to .zip, extract it using your operating system, and open the .kml file inside.

7.5.2 Exporting Drawings as GPX or KML

To export the drawings currently on the map, use File > Map > Export Drawings. Choose GPX or KML format. All lines and polygons are exported. Circles are not exported as neither format has an equivalent geometry type.

Exported GPX files can be loaded on Garmin GPS devices, opened in CalTopo, BaseCamp, or any GPX-compatible application. Exported KML files open in Google Earth, CalTopo, or any KML-compatible application.

 Tip

For SAR operations: draw your search segments in APRS Command, export as KML, and import into CalTopo to overlay on your probability maps. Or export your CalTopo assignment segments as GPX and import them into APRS Command so field operators can see their segments alongside live APRS positions.

7.6 CalTopo / SARTopo Integration

CalTopo (also known as SARTopo) is the mapping platform used by search and rescue teams across North America. APRS Command integrates with CalTopo in two ways: file-based exchange using GPX/KML, and live position forwarding that sends APRS station positions directly to your CalTopo map in real time.

7.6.1 File-Based Exchange (GPX and KML)

This workflow requires no configuration and works with a free CalTopo account.

CalTopo → APRS Command

1. In CalTopo, open your SAR map and click Export at the top of the Map Objects panel.
2. Choose Download GPX File or Download KML File.
3. In APRS Command, use File > Map > Import GPX/KML and select the downloaded file.

Your CalTopo assignment segments, clue locations, and search area boundaries now appear as overlays on the APRS map alongside the live positions of field operators. SAR coordinators get APRS situational awareness without leaving their CalTopo workflow.

APRS Command → CalTopo

1. In APRS Command, draw your search segments or planned routes using the Draw Line and Draw Polygon tools.
2. Use File > Map > Export Drawings and choose KML format.
3. In CalTopo, click Import in the Map Objects panel and select your KML file.

Alternatively, use this workflow after a search to import tracked field operator routes. Select operator tracks from the Packet Log, export as GPX, and import into CalTopo for post-mission probability analysis and ICS documentation.

7.6.2 Live Position Forwarding

When live forwarding is enabled, APRS Command automatically sends every received station position to your CalTopo map as it arrives. SAR coordinators watching CalTopo see APRS-tracked field operator positions updating in real time, overlaid on their assignment segments — without any manual export/import step.

Important

Live forwarding requires internet access. This feature is suitable for command post operations with a network connection. For fully offline field deployments, use the file-based exchange workflow described above.

Setting up CalTopo for Live Forwarding

1. Open your CalTopo map at caltopo.com.
2. Click + Add in the Map Objects panel.
3. Select Locator, then Live Track — Fleet, Email, Other.
4. Give the locator a name (e.g. "APRS Command Feed") and click Add.
5. Note the Map ID from the URL in your browser — it is the 4–6 character code after /m/ (example: caltopo.com/m/AB12 → Map ID is AB12).

Configuring APRS Command

1. Open Settings > Integrations > CalTopo.
2. Enter the Map ID you noted above.
3. Choose which sources to forward — APRS-IS packets, RF packets, or both.

4. Set the minimum update interval. The default is 60 seconds — each individual station is forwarded at most once per minute. Lower values increase CalTopo update frequency but generate more network traffic.
5. Enable the toggle and click Save.

Once enabled, every position packet received by APRS Command — from any station, on any connected source — is forwarded to your CalTopo map. The operator callsign becomes the pin label on the CalTopo map. Multiple field operators appear as separate labeled pins.

**Note**

No CalTopo account, API key, or authentication is required for live forwarding. The position reporting endpoint is publicly available. The only requirement is the Map ID of a CalTopo map that has a Locator object added to it.

**Tip**

For a SAR activation: enable live forwarding at the command post laptop running APRS Command. All field operators with APRS trackers appear on both the APRS Command map (with movement trails, messaging, and ICS documentation) and the CalTopo map (with assignment segments, clue locations, and probability areas) simultaneously. The two tools complement each other rather than competing.

7.7.1 Weather Stations

Shows weather readings from APRS weather stations in your area — temperature, wind speed and direction, barometric pressure, and rainfall. Click any weather station icon on the map to see its current data.

7.7.2 NWS Weather Alerts

Displays active National Weather Service alerts for your area — tornado warnings, severe thunderstorm warnings, winter storm advisories. A badge on the alert icon shows the count of active alerts. Alerts are color-coded by severity: deep red for Extreme, red for Severe, orange for Moderate, yellow for Minor.

7.7.3 Radar Overlay

See Section 6.10. Available from both the sidebar and Weather > Radar Overlay: ON/OFF.

7.7.4 Weather History

The Weather window includes a History tab showing a rolling chart of temperature, humidity, and wind speed from a selected weather station. APRS Command stores up to 144 readings per station (24 hours at 10-minute intervals). History accumulates while APRS Command is running and is not saved between sessions.

1. Open the Weather window from the menu bar.
2. Click the History tab.
3. Select a station from the dropdown.
4. Charts appear for temperature (°F), humidity (%), and wind speed (mph).

7.7.5 Connecting a Home Weather Station

APRS Command can receive live readings from a personal weather station and encode them into standard APRS weather packets. Seven integration methods are supported:

Method	Best For
Ecowitt / Fine Offset / GW1000	GW1000/GW2000 gateways and Fine Offset-based stations (Ambient Weather, Froggit, Misol, Sainlogic). Local network HTTP — no cloud required.
Davis WeatherLink (cloud)	Vantage Pro 2, Vantage Vue, and Envoy stations via WeatherLink Cloud API.
Ambient Weather (cloud)	WS-2902, WS-5000, and other Ambient Weather stations via Ambient Weather Network API.
WeatherFlow Tempest (cloud)	Tempest hub via WeatherFlow cloud API. Requires personal access token from tempestwx.com.
WeatherFlow Tempest (local UDP)	Tempest hub broadcasting locally on port 50222. No internet — ideal for field operations.
Peet Bros Ultimeter (serial)	Ultimeter 100, 800, 2000, 2100, and Pro series via direct serial connection.
Weather Software File Import	CumulusMX, WeeWX, Weather Display, or any software writing realtime.txt or similar files.
Manual Entry	Portable operations or one-time beacons where you enter readings by hand.

7.8 Tools

7.7.1 Offline Map Download

1. Open Tools > Offline Map Download.
2. Navigate the map to the area you want to cache.
3. Set the zoom level range.
4. Click Download.

Caution

Offline map downloads can be large — tens to hundreds of megabytes for a wide area. Ensure sufficient disk space before downloading.

7.7.2 Frequency Reference

A built-in database of amateur radio frequencies — repeaters and simplex channels. Browse, search, and add your own entries.

7.7.3 Elevation Profile

Shows terrain elevation between two points. Useful for understanding radio coverage.

1. Open Tools > Elevation Profile.

2. Click two points on the map.
3. The elevation chart shows height along the path with total distance.

**Tip**

A ridge visible in the elevation profile between two stations often explains why communications are difficult, even when the stations appear close on the map.

7.7.4 Shadow Beacon

Transmits a position beacon on behalf of a non-APRS station so they appear on the map.

1. Open Tools > Shadow Beacon.
2. Enter the station's callsign.
3. Enter their current position.
4. Set a beacon interval and click Start.

**Important**

Only use shadow beaconing with explicit permission from the station you are representing. Transmitting position data under another operator's callsign without their knowledge is improper operating practice.

7.8.5 Coverage Prediction

Displays a predicted coverage circle based on a station's PHG data (Power, Height, Gain).

1. Select a station on the map.
2. Open Tools > Coverage Prediction.
3. The estimated coverage area appears as a colored circle.

**Note**

Coverage prediction is an estimate based on simplified propagation models. Use it as a planning guide, not a guarantee.

7.7.6 Net Script Editor

Create pre-written message scripts with template variables:

- {{callsign}} — replaced with the selected station's callsign
- {{date}} — replaced with today's date
- {{time}} — replaced with the current time
- {{net_name}} — replaced with the net name you configure

7.7.7 Winlink RMS Gateways

Shows nearby Winlink Radio Message System gateway stations on the map. Winlink is used in emergency communications for email over radio.

**Note**

The Winlink gateway list requires an API key from Winlink.org. Enter it in Settings > Connections after receiving your key.

7.9 Packet Statistics Dashboard

The Packet Statistics Dashboard gives a live view of everything APRS Command has received since the session started. Opens from the menu bar. Updates automatically every two seconds.

7.8.1 The Four Headline Numbers

Counter	What It Means
Total Packets	Every APRS transmission received this session. Resets on Reset Statistics but not on temporary disconnections.
Unique Stations	Number of distinct callsigns heard. KE4CON and KE4CON-1 are counted separately. Grows throughout the session and never decreases.
Packets Per Hour	Running average since session start. Quiet rural: 20–50/hr. Busy urban: 200–500/hr. Active net: 1,000+/hr.
Packets Per Minute	Same average per minute. More useful during short sessions.

7.8.2 Packet Types Breakdown

Type	What It Represents
Position (blue)	Station location beacons. Typically 70–90% of all APRS traffic. Populates the map.
Message (green)	Direct messages between callsigns, including acknowledgments.
Weather (purple)	Meteorological readings from APRS weather stations.
Object / Item (amber)	Markers for locations or resources — not radio stations themselves.
Status (gray)	Brief text announcements without coordinates. Common during active nets.
Other / Unknown (slate)	Telemetry, capabilities announcements, and third-party forwarded traffic.
Invalid / Malformed	1–5% is normal. Above 10–15% may indicate RF interference or equipment issues.

7.8.3 Most Active Stations

Shows the top 15 callsigns by total packets received. The most active is often a digipeater or iGate — this is normal. During a net, use this panel to see which operators are active and spot any conspicuous absences.

7.8.4 Hourly Activity Chart

A bar chart showing packets received in each of the last 24 hours. Useful patterns:

- Morning commuter peak (7–9 AM) — mobile stations produce a noticeable increase.
- Net times — scheduled nets produce a sharp spike at their start time.
- Overnight quiet — most areas drop significantly midnight to 6 AM.

- Event activation — sustained high activity across multiple consecutive hours.

7.8.5 Reset Statistics

Clears all counts and restarts accumulation from zero without disconnecting. Useful at the start of a net or event for a clean baseline.

7.8.6 Aloha Circle

The Aloha circle — defined by Bob Bruninga WB4APR — estimates the effective range within which your station can reliably receive packets given the current channel load. As more stations transmit on the shared 144.390 MHz channel, collisions increase and effective range shrinks. The panel displays the calculated radius and current channel load percentage.

Note

The Aloha circle is an estimate based on observed traffic rate. Actual range also depends on your antenna, terrain, and individual station behavior.

7.10 Session Templates

Session Templates configure the entire application for a specific event type with a single click — beacon path, comment, interval, and filter radius all update immediately without a restart.

[Screenshot]

Session Templates window — template list and detail panel

7.9.1 What a Template Configures

Setting	What It Controls
Beacon path	The digipeater routing path (e.g. WIDE1-1,WIDE2-1 or WIDE1-1 for hilltop portable).
Beacon comment	Text appearing alongside your callsign on other operators' maps.
Beacon interval	How often your position is transmitted automatically, in minutes.
Filter radius	How far from your station APRS-IS sends you traffic.

Note

Templates never change your callsign, SSID, location, passcode, or any other setting. It is always safe to apply a template at any time.

7.9.2 Built-In Templates

Template	Designed For	Key Settings
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Public Service Event	Marathons, parades, bike rides, and community events.	5-min interval, 50-mile filter, comment: "Public Service"
Emergency Communications	ARES/RACES emergency activations supporting a served agency.	5-min interval, 100-mile filter, comment: "EMCOMM ACTIVATED"
Weekly Check-In Net	Regular scheduled nets with roll call and announcements.	10-min interval, 75-mile filter, comment: "Net Control"
SOTA / POTA Portable	Hilltop or park activations. Single-hop path prevents congestion from high-altitude sites.	10-min interval, 200-mile filter, path: WIDE1-1
Skywarn Weather Spotter	NWS Skywarn activations for severe weather spotting.	5-min interval, 100-mile filter, comment: "Skywarn Spotter Active"

7.9.3 Creating a Custom Template

1. Open Session Templates from the menu bar.
2. Click + New Custom at the bottom of the template list.
3. Enter a Template name and Description.
4. Fill in the settings you want the template to apply.
5. Add Operator notes — a checklist or reminder visible when this template is selected.
6. Click Save Template, or Apply Now to save and apply immediately.

To edit a custom template, select it and modify the fields. To delete it, click the Delete button. Built-in templates cannot be deleted.

7.9.4 Applying a Template

1. Open Session Templates from the menu bar.
2. Select the template from the list on the left.
3. Review the settings preview and operator notes.
4. Click Apply This Template. Settings take effect immediately.

7.11 Voice Readout

Uses your operating system's built-in text-to-speech engine to announce APRS events aloud. Requires no internet connection, no API key, and no additional software on macOS or Windows.

7.10.1 Platform Support

Platform	TTS Engine	Setup Required?
macOS	Built-in 'say' command. Ships with many voices.	None. Additional voices: System Settings > Accessibility > Spoken Content.

Windows 11	System.Speech built into Windows.	None. Additional voices: Settings > Time and Language > Speech > Add voices.
Linux / Raspberry Pi	espeak-ng	Install: <code>sudo apt install espeak-ng</code> . APRS Command will prompt if not found.

7.10.2 Alert Types

Alert Type	What Triggers It	Default
Incoming Messages	Direct APRS message arrives addressed to your callsign. Speaks sender and message body.	On
Net Check-Ins	Station checks in to your net. Speaks the callsign.	On
NWS Weather Alerts	National Weather Service alert arrives. Speaks the alert type and area.	On
Station Alerts	A tracked station appears, moves, or triggers a geofence. Can be noisy in busy areas.	Off
Connection Events	APRS-IS connects or disconnects. Useful when the app runs in the background.	Off
Beacon Confirmations	Each successful beacon transmission is announced.	Off

Voice Readout runs on a background thread and never blocks the user interface. If the queue fills during a very busy activation, the oldest waiting alert is dropped to prevent a backlog.

7.12 Events — ICS Documentation

The Events menu generates standard ICS (Incident Command System) documentation from activity logged during your session.

ICS Form	Contents
ICS-205 Radio Communications Plan	Frequencies, callsigns, and assignments.
ICS-211 Check-In List	All stations that checked in during the session.
ICS-213 General Message Form	One form per APRS message sent.
ICS-214 Activity Log	Chronological log of significant events.

1. Open Events > After-Action Report.
2. Review the session activity summary.
3. Select which ICS forms to include.
4. Click Export.

7.13 Telemetry Monitor

Displays live APRS telemetry packets (type T#) as they arrive. Telemetry packets carry up to five analog values (A1–A5) and eight digital bit values from remote sensors, repeater controllers, and other automated stations.

Column	What It Shows
Time	When the packet was received (local time, HH:mm:ss).
Callsign	The station that transmitted the telemetry.
Seq	The sequence number (0–999, rolling).
A1–A5	The five analog channel values as decoded integers.
Digital	Eight digital bits shown as a binary string, e.g. 00000000.
Raw	The raw telemetry body from the APRS packet.

Click any row to see the full detail panel. Use the callsign filter or station dropdown to focus on a specific station. The Clear button removes all records. Up to 500 packets are retained at a time.



Note

The meaning of each analog channel is defined by PARM, UNIT, and EQNS metadata packets from the same station. APRS Command displays raw decoded values — interpretation requires knowing the station's metadata.

7.14 Scheduled Beacons

Configure time-of-day triggers that automatically send a beacon at a specific time on specific days of the week. Useful for regular check-ins, net participation, or any situation where you want APRS Command to beacon predictably without manual intervention.

Field	What It Controls
Label	A descriptive name, e.g. "Monday morning net".
Time	The hour (0–23) and minute (0–59) when the beacon fires.
Days	Which days of the week the trigger is active. Check Mon–Sun individually.
Custom comment	Optional text overriding your station comment for this beacon only.
Enabled	Uncheck to temporarily disable without deleting.



Note

APRS Command must be running at the scheduled time for the beacon to fire. The same safety checks as manual beacons apply — if transmit is inhibited or APRS-IS is disconnected, the beacon is skipped.

7.15 Keyboard Shortcuts

A complete reference is available at any time: press Ctrl+/ or open Keyboard Shortcuts from the menu.

7.14.1 Global Shortcuts

Shortcut	Action
Ctrl+N	Open Net Control window
Ctrl+B	Beacon Now — send an immediate position beacon
Ctrl+D	Toggle Dark / Light mode
Ctrl+/	Open the Keyboard Shortcuts reference window

7.16 Dead Reckoning

Dead reckoning projects a mobile station's current position forward in time based on its last known course and speed. APRS Command calculates this automatically for qualifying stations and shows the estimate in the station detail panel when you click the station on the map.

Dead reckoning applies when all of the following are true:

- The station's last packet included a course (degrees true) and speed (knots).
- Speed is at least 1 knot — stationary stations are not projected.
- The last packet was received within the last 30 minutes.

When applicable, a blue estimate panel appears in the station detail panel showing the projected distance traveled since the last fix. The map marker stays at the last confirmed position — dead reckoning is clearly labeled as an estimate. The projection distance respects your preferred distance unit.

Note

Dead reckoning becomes less accurate as time passes and as the station changes speed or direction. It is an estimate based on the last known heading, not a GPS fix.

7.17 Software Updates

APRS Command checks for updates automatically about 15 seconds after startup. The check is completely silent unless a newer version is found — if you are operating off-grid with no internet connection, nothing appears and nothing is affected.

When an update is available, a notification appears in the corner of the main window:

- If you installed APRS Command with an installer, clicking the notification downloads the update and restarts the app into the new version automatically. Your settings are preserved.
- If you are running a portable (zip) copy, clicking the notification opens the GitHub releases page in your browser so you can download the new version.

You can also check manually at any time: open About APRS Command from the menu bar, or visit github.com/KE4CON/APRS-Command/releases and compare the latest version against the one shown in the About box.

 Note

The automatic check never blocks startup and never shows errors. No internet simply means no check — ideal for field deployments.

7.18 Mobile Companion Web View

The Mobile Companion lets you monitor your APRS Command station from a phone or tablet on the same network. It serves a lightweight web dashboard showing your connection status, packet statistics, top stations, recent messages, and the net control roster.

1. Open Tools > Mobile Companion Web View from the menu bar.
2. A window appears showing a URL such as `http://192.168.1.50:52134/`
3. On your phone, connect to the same WiFi network as the computer running APRS Command.
4. Open the phone's web browser and enter the URL exactly as shown.
5. The dashboard loads and refreshes automatically while APRS Command is running.

 Important

The Mobile Companion has no password and uses unencrypted HTTP. Anyone on the same network can open the dashboard if they know the address. This is fine on your home network or a private field network you control — but avoid using it on public WiFi (hotels, coffee shops, event venues with open networks), where strangers share the same network.

 Note

The dashboard is read-only — it displays information but cannot transmit, change settings, or control the station. Firewall software on your computer may ask permission the first time the companion starts; allow access on private networks.

Chapter 8

RF and TNC Connection

APRS Command can receive and transmit APRS packets directly over radio using a hardware TNC or software modem. RF operation lets you participate in local activity not on APRS-IS, operate without internet, and run as an iGate. All connection paths can be active simultaneously.

8.1 The Four Connection Paths

Path	Description	Best For
APRS-IS	Internet connection to the global APRS network. The default — requires no radio or TNC.	Beginners. Fixed home stations. Monitoring and messaging.
KISS-TCP	Network connection to a KISS-capable TNC or software modem (GrayWolf, Direwolf).	GrayWolf field server deployment. Direwolf on a separate Pi.
Serial KISS	Direct serial port connection to a hardware TNC.	Kantronics KPC-3+, TNC-Pi, Mobilinkd, DigiRig, Kenwood TM-D710.
AGWPE (TCP)	Connection to AGW Packet Engine or BPQ32. Familiar to UI-View32 users.	Existing AGWPE or BPQ32 installations. Multi-port digipeaters.

8.2 Hardware TNC (Serial KISS)

8.2.1 Finding Your Port Name

Platform	How to Find the Port Name
macOS	Open Terminal and run: <code>ls /dev/tty.*</code> — appears as <code>/dev/tty.usbserial-XXXX</code> or <code>/dev/tty.usbmodem-XXXX</code> .
Windows	Open Device Manager > Ports (COM and LPT). Appears as COM3, COM4, or similar.
Linux / Raspberry Pi	Run: <code>ls /dev/ttyUSB* /dev/ttyACM*</code> — typically <code>/dev/ttyUSB0</code> or <code>/dev/ttyACM0</code> . If you get a permission error: <code>sudo usermod -aG dialout \$USER</code> (then log out and back in).

8.2.2 Baud Rate Reference

TNC / Device	Serial Baud Rate
Kantronics KPC-3+	9600 (default)
TNC-Pi (original and v2)	9600 (default)

Mobilinkd TNC3 / TNC4	38400
DigiRig in KISS mode	9600 (default)
Kenwood TM-D710 internal TNC	9600
Most other hardware TNCs	9600

8.2.3 Adding a Serial KISS Port

1. Open Settings > Connections.
2. Click + Add Port.
3. Set Type to Serial KISS (hardware TNC).
4. Enter the Port Name from Section 8.2.1.
5. Set the Baud Rate from Section 8.2.2.
6. Check Enabled and Receive Enabled. Check Transmit Enabled only if you want APRS Command to key up your radio.
7. Click Save. APRS Command connects immediately — no restart required.

8.3 KISS-TCP (GrayWolf / Direwolf)

Use KISS-TCP when your TNC or software modem runs as a network service. This is the recommended path for GrayWolf field deployment.

GrayWolf (by Chris Snell NW5W) is a single-binary Rust-based APRS station for headless Raspberry Pi operation. It exposes a KISS-TCP interface that APRS Command connects to as a client.

1. Ensure GrayWolf is running on the Pi with KISS-TCP enabled (default port 8001).
2. Find the Pi's IP address on your local network.
3. In APRS Command open Settings > Connections and click + Add Port.
4. Set Type to KISS TCP. Enter the Pi's IP address and port 8001.
5. Check Enabled, Receive Enabled, and Transmit Enabled.
6. Click Save. APRS Command connects and begins receiving RF packets.



Tip

If GrayWolf runs on the same computer as APRS Command, use 127.0.0.1 as the host instead of a network IP address.

8.4 AGWPE and BPQ32

APRS Command supports AGW Packet Engine (AGWPE) and BPQ32 — the same connection path used historically by UI-View32, making it familiar to many existing operators.

8.4.1 Setting Up the Connection

AGWPE and BPQ32 both listen on TCP port 8000 by default.

1. Start AGWPE or BPQ32 and verify it is connected to your TNC.
2. In APRS Command, go to Settings > Connections and click + Add Port.

3. Select AGWPE (TCP) as the port type.
4. Enter the host: 127.0.0.1 for a local installation, or the IP address for a remote host.
5. Set port to 8000 (or your configured port if changed).
6. Check Enabled and Receive Enabled.
7. Click Save. APRS Command connects automatically.

**Note**

AGWPE must be running before APRS Command connects. If not running, APRS Command retries every 10 seconds.

8.4.2 BPQ32 on Raspberry Pi

BPQ32 on a Raspberry Pi can serve as a combined node, digipeater, and APRS gateway. Connect over the local network using the AGWPE interface. Verify BPQ32 has the AGWPE interface enabled in BPQ32.cfg.

**Tip**

For GrayWolf-based field servers, prefer the KISS-TCP path (Section 8.3) — GrayWolf's native KISS-TCP interface is more efficient for APRS use than the AGWPE layer.

8.5 Transmit Settings

APRS Command will not key up your radio unless all three of the following are enabled:

- Transmit Enabled (Settings > Connections, per port) — master switch for this port.
- RF Transmit Enabled (Settings > Station Setup) — global RF transmit toggle.
- Valid callsign configured in Station Setup.

8.6 Path Selection Guide

Path	When to Use It
WIDE1-1,WIDE2-1	Standard for most fixed and mobile operations. Two hops — fill-in then wide-area digipeaters. Good regional coverage.
WIDE1-1	Single fill-in hop. For hilltop portable (SOTA, POTA) where your signal is already strong. Prevents flooding the network.
WIDE2-1	One wide-area hop. For areas with good wide-area coverage but no fill-in digipeaters.
(blank/empty)	Direct only — not relayed. For local bench testing only.

8.7 iGate Operation

An iGate receives packets from RF and forwards them to APRS-IS. Running an iGate with a good antenna is one of the most valuable contributions an operator can make to the local APRS network.

1. Open the iGate panel from the sidebar.

2. Check Enable iGating and RF to APRS-IS Gating.
3. Click Save. The iGate status panel shows gating activity in real time.

8.8 Verifying RF Operation

The Raw Packet Monitor (menu bar) shows every packet received and transmitted on every active port in real time. Open it and watch for incoming packets — if packets appear, your TNC is receiving. To verify transmit, use Beacon Now, then check aprs.fi for your callsign.

8.9 Troubleshooting RF

No packets in the Raw Monitor

- Check that the port is Enabled and Receive Enabled in Settings > Connections.
- Verify the port name or IP address is correct. On macOS and Linux, the serial port name is case-sensitive.
- Confirm the TNC is in KISS mode — many TNCs start in command mode.
- Try a different baud rate.
- On Linux/Pi, check: `ls -la /dev/ttyUSB0`. If the group is dialout, run: `sudo usermod -aG dialout $USER` and log out and back in.
- Ensure your radio is on 144.390 MHz (VHF APRS in North America).

Port already in use

Another application has the serial port open. Common culprits: a terminal emulator, another APRS client, or on Linux the ModemManager service. To stop ModemManager: `sudo systemctl stop ModemManager`

Transmit not working

- Confirm Transmit Enabled is checked for the port in Settings > Connections.
- Confirm RF Transmit Enabled is checked in Settings > Station Setup.
- Verify PTT wiring between your TNC and radio.
- Check the Raw Packet Monitor for packets marked direction OUT. If they appear, APRS Command transmitted — a radio not keying up points to a PTT wiring issue.

Chapter 9

Map Interaction

9.1 Navigating the Map

Action	How to Do It
Pan	Click and drag with the left mouse button.
Zoom in	Scroll mouse wheel up.
Zoom out	Scroll mouse wheel down.
Reset view	Click the Home icon (🏠) in the sidebar.
Center on station	Double-click a station icon, or use Find Station (🔍).

9.2 Clicking Stations

Click any station icon to select it. A details panel appears showing:

- Callsign and SSID
- Last heard time and age
- Position (latitude/longitude and Maidenhead grid square)
- Distance and bearing from your station in your preferred unit
- Symbol and comment
- Speed and course (for mobile stations)
- Dead reckoning estimate when applicable (see Section 7.15)
- Digipeater path and heard-by information
- Raw packet data

From the details panel you can send a direct message to that station, or set range rings centered on their position.

[Screenshot]

Station details panel — all fields visible

9.3 Right-Click Context Menu

Right-clicking anywhere on the map opens a context menu:

Menu Item	What It Does
Create Object Here	Places a new APRS object at the clicked location with coordinates pre-

	filled.
Set Ring Center	Moves range rings to be centered on this point.
Measure from Here	Starts a distance measurement from this point.
Elevation Profile from Here	Starts an elevation profile path from this point.
Copy Coordinates	Copies the latitude/longitude to the clipboard in decimal-degree format.

Chapter 10

Troubleshooting

Problems are listed by symptom — what you see — not by technical cause. Work through each step in order before moving to the next.

10.1 No Stations Appearing on the Map

1. Check the title bar. Is the APRS-IS indicator green? If not, you are not connected — check your internet connection.
2. Open Settings > Connections. Confirm server is rotate.aprs2.net and port is 14580.
3. Check your APRS-IS filter radius in Station Setup. If very small, few stations may fall within it. Try temporarily increasing it.
4. Verify your latitude and longitude in Station Setup. An incorrect position places your filter in the wrong part of the world.

**Note**

APRS activity varies significantly by location and time of day. Verify by checking aprs.fi in a web browser to see what is actually active in your area.

10.2 Beacons Errors

"APRS-IS Beacons is Disabled"

1. Open Settings > Station Setup.
2. Confirm Enable Transmit is checked.
3. Confirm Enable APRS-IS Transmit is checked.
4. Click Save, then try Beacon Now again.

"Beacon Interval Too Short"

The minimum beacon interval is 5 minutes. Open Settings > Station Setup > APRS-IS Beacon Interval and set it to 5 or higher. Click Save.

"APRS-IS Client Not Connected"

1. Open Settings > Connections.
2. Confirm a valid passcode is entered — not -1. Find yours at apps.magicbug.co.uk/passcode.
3. Click Save and wait about 10 seconds for the transmit connection to establish.
4. Try Beacon Now again.

"Beacon Transmitted" but not on aprs.fi

- Spaces in Beacon Path — open Settings > Station Setup > Beacon Path. It must read exactly WIDE1-1,WIDE2-1 with no spaces. Click Save to trigger automatic correction.

- Authentication not complete — if you clicked Beacon Now within a second of saving settings, the server login handshake may not have completed. Wait 10 seconds, then beacon again.

10.3 App Crashes or Won't Start

1. Close the app completely.
2. Check the error log:
 - macOS: ~/Library/Application Support/APRS Command/logs/startup-error.log
 - Windows: %APPDATA%\APRS Command\logs\startup-error.log
 - Linux/Pi: ~/.config/APRS Command/logs/startup-error.log
3. Include the error log when reporting the issue at github.com/KE4CON/APRS-Command/issues.

Tip

Run the app from a Terminal or Command Prompt during testing — many errors print useful messages there even when nothing appears in the log file.

10.4 Radar Not Appearing

1. Confirm internet connection is active.
2. Wait up to 30 seconds — NOAA's server can be slow to respond.
3. Toggle radar off and back on.
4. If the animation scrubber never appears after 30 seconds, NOAA's service may be temporarily unavailable. Try again later.

10.5 Resetting to Default Settings

Important

This deletes all saved settings including your callsign, location, and connection configuration. Write these down before proceeding.

- macOS: delete ~/Library/Application Support/APRS Command/
- Windows: delete %APPDATA%\APRS Command\
- Linux/Pi: delete ~/.config/APRS Command/

10.6 Getting Help

- Check github.com/KE4CON/APRS-Command/issues for known issues and workarounds.
- Open a new issue including: your operating system, exact error message, what you were doing, and the error log contents.

Chapter 11

Developer Integration

APRS Command includes a comprehensive developer integration system allowing external programs, scripts, and tools to connect to a running instance — reading data, submitting data, receiving live events, and requesting transmissions.

Three Integration Paths

Path	How It Works	Best For
Local REST API (port 8765)	HTTP request/response on localhost. 21 endpoints for reading and writing station, weather, message, object, GPS, alert, and diagnostic data.	Polling queries. One-shot data retrieval. Scripts.
WebSocket Event Streams (port 8766)	Push-based live event delivery. Nine topic streams: stations, weather, objects, messages, alerts, raw packets, RF diagnostics, system events, and combined.	Real-time displays. Live logging. Alerting systems.
File Hook System	Folder-based data exchange requiring no network connection. APRS Command exports JSON snapshots on a schedule; external programs drop import files in a watched folder.	Offline tools. Tools that cannot open TCP connections.

Enabling Developer Integration

1. Open Settings > Developer.
2. Enable the Local REST API, WebSocket Event Streams, and/or File Hooks as needed.
3. Set an API token for the REST API and WebSocket streams (strongly recommended).
4. Configure permissions for each integration path.

All paths enforce a seven-tier permission model. The default is `ReadOnly` — external programs can read all data but cannot write or transmit. Transmit permissions must be explicitly granted by the operator.

Important

Transmit permissions (`TransmitAprsIs` and `TransmitRf`) allow external programs to key up your radio. Grant these only to trusted, reviewed software.

Full documentation is in the APRS Command Developer Manual, available at github.com/KE4CON/APRS-Command.

Appendix A — Common APRS Symbols

Table	Code	Symbol Description
/	-	House / Home station
/	>	Car / Vehicle
/	j	Jeep
/	R	Recreational vehicle (RV)
/	Y	Yacht / Boat
/	k	Truck
/	f	Fire truck
/	u	Bus
/	*	Snowflake / Weather station
/	_	Weather station (WX packet format)
/	r	Antenna / Radio tower
/	I	TCP/IP node
/	D	Digipeater
/	S	Satellite
/	O	Balloon
\	I	Area object (shape drawn on map)

Note

A complete APRS symbol reference is available at aprs.org/symbols.html

Appendix B — Beacon Path Reference

Path	When to Use It
WIDE1-1,WIDE2-1	Standard for most fixed and mobile stations. Fill-in then wide-area digipeaters.

	Good regional coverage without excessive congestion.
WIDE1-1	Single fill-in hop. For hilltop portable (SOTA/POTA) where your signal is already strong. Prevents flooding the network from a high-altitude site.
WIDE2-1	One wide-area hop. For areas with good wide-area coverage but no fill-in digipeaters.
NOGATE	Add this keyword to prevent your packet from being gated to APRS-IS. RF-only operation.
(blank/empty)	Direct only — not relayed by any digipeater. For bench testing only.

Important

Never put spaces in your beacon path. WIDE1-1,WIDE2-1 is correct. WIDE 1-1,WIDE 2-1 will be silently rejected by APRS-IS.

Appendix C — Glossary

Common APRS and amateur radio terms used throughout this manual.

AGWPE

AGW Packet Engine — a protocol and software application for connecting packet radio software to a TNC or sound card modem.

APRS

Automatic Packet Reporting System. A real-time digital communication system for position, message, weather, and status data.

APRS-IS

The APRS Internet Service network. Carries APRS packets over the internet, connecting the global APRS network.

Beacon

A periodic packet transmitted by a station announcing its position, status, or other information.

BPQ32

Packet radio node and BBS software by John Wiseman G8BPQ with an AGWPE-compatible interface.

Callsign

The licensed station identifier assigned by a licensing authority. Example: KE4CON.

Digipeater

A station that receives APRS packets and re-transmits them on the same frequency, extending range.

Filter

An APRS-IS server-side filter limiting which packets the server sends to your client.

Geofence

A defined geographic area (circle, polygon, or rectangle) used to trigger an alert when a tracked station enters or exits.

GPSD

A Linux/Unix service (GPS Daemon) providing GPS data to multiple applications over a local network interface.

GrayWolf

A single-binary Rust-based APRS station by Chris Snell NW5W, designed for headless Raspberry Pi field server operation.

iGate

A station moving packets between the RF APRS network and APRS-IS. Bi-directional iGates also inject internet packets onto the local RF network.

ICS

Incident Command System. The standardized management system used by emergency management agencies.

KISS

Keep It Simple, Stupid. A minimal protocol for moving packet data between software and a TNC.

NMEA

A standard text sentence format output by GPS receivers.

Object

An APRS marker representing any location or resource, created and transmitted by an operator.

Packet

A single APRS transmission containing a position report, message, weather reading, object, or status update.

Passcode

A number derived from your callsign that authenticates your station on APRS-IS for transmitting. Keep it private.

Path

The routing instructions in an APRS packet that tell digipeaters how many times to relay it.

PHG

Power, Height, Gain. A standard APRS field describing a station's antenna system for coverage estimation.

RF

Radio Frequency. RF operation means transmitting and receiving over the air using a radio and TNC.

SSID

Secondary Station Identifier. A number 0-15 appended to a callsign to distinguish multiple stations. KE4CON-1 has SSID 1.

Symbol

A two-character code defining the icon displayed for a station or object on the map.

TNC

Terminal Node Controller. Hardware (or software) converting between packet data and radio/audio signals.

UI-View32

A widely used APRS client by Roger Barker G4IDE. Closed source; source code destroyed on Roger's passing in 2004. APRS Command is a spiritual successor, built open-source so the community can carry it forward indefinitely.

Winlink

A worldwide radio email system used in emergency communications. APRS Command includes a Winlink RMS Gateway locator.

APRS Command · github.com/KE4CON/APRS-Command · GPL v3 · 73 de KE4CON